

**COURSE TITLE : INDUSTRIAL ELECTRONICS AND CONTROL DRIVES**  
**COURSE CODE : 4212**  
**COURSE CATEGORY : A**  
**PERIODS/WEEK : 4**  
**PERIODS/SEMESTER: 56**  
**CREDITS : 4**

#### TIME SCHEDULE

| Module       | Topics                      | Period    |
|--------------|-----------------------------|-----------|
| 1            | Power semiconductor devices | 14        |
| 2            | Converters and inverters    | 14        |
| 3            | AC&DC Motors                | 14        |
| 4            | Control drives              | 14        |
| <b>TOTAL</b> |                             | <b>56</b> |

#### COURSE OUTCOME

| Module | G.O. | On completion of the study of this module the student will be able        |
|--------|------|---------------------------------------------------------------------------|
| 1      | 1    | To understand the working and application of power semiconductor devices. |
| 2      | 1    | To understand the working and application of SCR in switching circuits    |
| 3      | 1    | To understand the working principle of DC & AC Motors                     |
| 4      | 1    | To understand single phase and three phase dual converters with waveforms |

On completion of the study the student will be able

#### MODULE I POWER SEMICONDUCTOR DEVICES

##### **1.1.0 To understand the working and application of power semiconductor device. Understand working and application of power devices**

- 1.1.1 To define Power diode and explain the structure of it.
- 1.1.2 To explain the circuit diagram and V-I characteristics ..of the power diode
- 1.1.3 To describe the structure and working principle of a SCR and the static characteristics.
- 1.1.4 To state the method of turn on (R,RC and UJT triggering) of SCR.
- 1.1.5 To define the terms, holding current, latching current gate, triggering current, turn on time and turn off time.
- 1.1.6 To study transistor analogy .
- 1.1.7 To describe the structure and working principle of a Triac and the static characteristics.
- 1.1.8 To describe the structure and working principle of a Diac and the static characteristics.

- 1.1.9 To explain the protection methods of switching devices
- 1.1.10 To explain the role of snubber circuits

## **MODULE II CONVERTERS AND INVERTERS**

### **2.1.0 To understand the working and application of SCR in switching circuits**

- 2.1.1 To explain various commutation techniques of SCR – brief description only
- 2.1.2 To understand the operation of single-phase half wave converter, full wave midpoint and bridge converters. Expression for output voltage
- 2.1.3 To understand the operation of three phase converter<sup>2</sup>
- 2.1.4 To understand the working of Triac light dimming circuit
- 2.1.5 To explain single phase series, bridge inverter with relevant waveforms and mathematical expressions
- 2.1.6 To understand switching sequence in bridge inverter, timing waveforms
- 2.1.7 To study the working principle of three phase bridge inverter.
- 2.1.8 To study static transfer switch
- 2.1.9 To study PWM Chip-SG3524

## **MODULE III AC&DC MOTORS**

### **3.1.0 To know the working principle of DC&AC motors**

- 3.1.1 To study the principle of operation of DC motors
- 3.1.2 To study Fleming's left hand rule
- 3.1.3 To study the shunt, series, compound motors
- 3.1.4 To list the factors affecting the speed of DC motors
- 3.1.5 To explain the principle of operations of single phase induction motor, Three phase induction motor universal motor, stepper motor
- 3.1.6 To study the working principle of AC and DC servo motors
- 3.1.7 To list the applications of servo motors
- 3.1.8 To understand the methods of speed control of induction motors
- 3.1.9 To describe variable voltage and variable frequency speed control
- 3.1.10 To explain soft start of AC and DC
- 3.1.11 To explain the working principle of AC&DC tachogenerator

## **MODULE IV CONTROL DRIVES**

### **4.1.0 To know single phase and three phase dual converters using waveforms**

- 4.1.1 To mention few application – 1-  $\phi$  & 3 –  $\phi$  dual converters
- 4.1.2 To understand the principles and applications of cyclo converters
- 4.1.3 To state the principle of choppers
- 4.1.4 To describe step up and step down choppers
- 4.1.5 To understand the working of Jone's chopper with waveforms
- 4.1.6 To study AC choppers
- 4.1.7 To mention the applications of chopper
- 4.1.8 To compare AC and DC drives
- 4.1.9 To state the requirements of variable speed drive
- 4.1.10 To explain the speed control of DC drive

## **CONTENT DETAILS**

### **MODULE I**

Power Devices-applications - Power diode - structure and VI characteristics - SCR – constructional features – operation – transistor analogy – characteristics – specifications –holding current – latching current – gate current - turn on time – turn off time - TRIAC – structure, working, static characteristics -DIAC – structure, working, static characteristics - Protection devices – snubber circuits

### **MODULE II**

Triggering and Commutation of SCR - Triggering of SCR - different methods of turn on – R triggering - RC triggering - triggering using UJT relaxation oscillator - Commutation of SCRs - Line commutation and forced commutation - different methods of forced - Commutation- schemes Class A, B, C, D, E, F-converters- Operation of single-phase half wave- full wave midpoint and bridge converters with -resistive and inductive loads, expression for output voltage -Three phase converters-Triac light dimming circuit-Inverters – Pulse width modulation (PWM) – series and parallel inverter - Single-phase series -inverter - single-phase bridge inverter (explanation with relevant wave forms and mathematical expressions) - Three phase bridge inverters- static transfer switch - PWM Chip-SG3524

### **MODULE III**

Principle of operations of DC motor – Fleming’s left hand rule – back emf - torque equation -load characteristics of shunt and series motors – single phase induction motor – Three phase induction motor Universal motor- application of single phase Induction motor –methods of speed control of induction motors-variable voltage-variable frequency- stepper motors – principle –applications – servo motors-tachogenerators -principle-operation –application - Soft start of AC and DC motors

### **MODULE IV**

Dual converters -Single phase and three phase dual converters- principle and operation-waveforms - Application-Cycloconverter for single phase – applications Choppers- Methods of output control-principle of step up and step down chopper- Jone’s chopper principle and working-waveforms AC chopper-application of choppers - Definition of drives – comparison between AC and DC drive – Requirement of variable speed drive-Speed control of DC drives

## **REFERENCES**

1. B.L.Theraja. A Text Book of Electrical Eng volume 2
2. M.H .Rashid, Power Electronics Circuits and Devices
3. Biswanath Paul, Industrial Electronics and Control (PHI)
4. Bimal.K.Bose, Modern Power Electronics and AC Drives
5. Joseph Vithayathil, Power Electronics Principles and Applications
6. S.K.Bhattacharya, S.chaterjee, Industrial Electronics and Control